THE FIBRE-DIMENSION BOUND AND THE SEMISMALLNESS OF THE PROJECTIVE MORPHISM BETWEEN TWO QUIVER VARIETIES

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Allowed standard prerequisites (editorial).

- The morphism pi from M_zeta(V,W) to M_{zeta^bullet}(V,W), the strata (M_{zeta^bullet})_(G-hat) and the chamber condition making M_zeta(V,W) nonsingular. Set up earlier in the same subsection (source lines 1324-1552, the author's subsections on strata and on the local structure). Fixes the map, its target stratification and the smoothness assumption under which the fibre-dimension bound is stated.
- Semismallness of a morphism with respect to a stratification. Used in part (2) of the statement (source lines 1566-1574). Names the property that part (2) establishes for the restricted map pi into its image.

Source reference. Original theorem 2.6: sources/algebra-fl01-quiver-varieties-and-branching/pinned-original-source.tex, source lines 944–944. Navigation pointer only: part (1) of the counted proof invokes the author's theorem that a certain moduli space is an affine algebraic manifold, in one clause; its statement is not reproduced here and the cardinality-free content it supplies is declared as a named standard prerequisite.

Source reference. Original equation 1.9: sources/algebra-fl01-quiver-varieties-and-branching/pinned-original-source.tex, source lines 660–660. Navigation pointer only: it labels the author's displayed trivial factor of the space bM, which the counted

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proof mentions in one clause; the stub keeps the author's own equation number resolvable without reproducing that display.

Source reference. Original equation 2.14: sources/algebra-fl01-quiver-varieties-and-branching/pinned-original-source.tex, source lines 1494–1494. Navigation pointer only: it labels the author's commutative diagram for the local isomorphism, which the counted proof names before drawing its conclusion; the diagram itself is not reproduced.

Proposition 0.1 (cf. [4, 6.11], [6, 10.4, 10.11])**.** *Let ζ, $\zeta^\bullet$ be as above and consider $\pi\colon \mathfrak{M}_\zeta(V,W) \to \mathfrak{M}_{\zeta^\bullet}(V,W)$. We further assume that ζ is in a chamber so that $\mathfrak{M}_\zeta(V,W)$ is nonsingular.*

(1) *Take a point x in a stratum $(\mathfrak{M}_{\zeta^\bullet})_{(\widehat{G})}$. If $\pi^{-1}(x) \neq \varnothing$, then we have*

$$\dim \pi^{-1}(x) \le \tfrac{1}{2}\big(\dim \mathfrak{M}_\zeta(V,W) - \dim(\mathfrak{M}_{\zeta^\bullet})_{(\widehat{G})}\big).$$

(2) *Replace the target of π by the image, hence consider $\pi\colon \mathfrak{M}_\zeta(V,W) \to \pi(\mathfrak{M}_\zeta(V,W))$. It is semismall with respect to the stratification $\pi(\mathfrak{M}_\zeta(V,W)) = \bigsqcup(\mathfrak{M}_{\zeta^\bullet})_{(\widehat{G})}$ where $\widehat{G}$ runs over the conjugacy classes of subgroups of G_V such that $(\mathfrak{M}_{\zeta^\bullet})_{(\widehat{G})}$ is contained in $\pi(\mathfrak{M}_\zeta(V,W))$.*

Proof. (1) This was proved under the assumption $\mathfrak{M}^{\mathrm{s}}_{\zeta^\bullet}(V,W) \neq \varnothing$ in [4, 6.11], but its proof actually gives the above. The point of the proof was that the fiber $\widehat{\pi}^{-1}(0)$ is a subvariety in $\widehat{\mathfrak{M}}_\zeta(\hat{V},\hat{W})$ (divided by the trivial factor (1.9)), which is an affine algebraic manifold by Theorem 2.6.

(2) By (2.14) and the subsequent remark, for each stratum $(\mathfrak{M}_{\zeta^\bullet})_{(\widehat{G})}$, the restriction of π to the preimage $\pi^{-1}((\mathfrak{M}_{\zeta^\bullet})_{(\widehat{G})})$ is a fiber bundle. Moreover, by (1), there exists a stratum $(\mathfrak{M}_{\zeta^\bullet})_{(\widehat{G})}$ such that $\dim(\mathfrak{M}_{\zeta^\bullet})_{(\widehat{G})} = \dim \mathfrak{M}_\zeta(V,W)$. Therefore we have $\dim \pi(\mathfrak{M}_\zeta(V,W)) = \dim \mathfrak{M}_\zeta(V,W)$. Hence (1) implies the assertion. □

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